2019/2020

① $z_1 = -i \;,\; z_2 = 2 + i\sqrt{3}$

a) $(z_1)^2 = (-i)^2 = -1$ B1

$\overline{z_2} = 2 - \sqrt{3}\, i$ B1

b) $W = \dfrac{-1 + 2 - \sqrt{3}\, i}{-i}$

$= \dfrac{1-\sqrt{3}\, i}{-i} \times \dfrac{i}{i}$ K1 conjugate

$= \dfrac{i - \sqrt{3}\, i^2}{1}$ K1 $i^2 = -1$

$= \sqrt{3} + i$ J1

$|W| = \sqrt{(\sqrt{3})^2 + 1^2}$ K1

$= 2$ J1

$\arg W = \theta = \tan^{-1} \left| \dfrac{1}{\sqrt{3}} \right|$ K1

$= \dfrac{\pi}{6}$ J1

② a) $3(5^{2x}) + 25^{\frac{1}{2}x+1} = 200$

$3(5^{2x}) + (5^2)^{\frac{1}{2}x+1} = 200$ K1 rule

$3(5^{2x}) + 5^{x+2} = 200$

$3(5^{2x}) + 25(5^x) - 200 = 0$ K1 quadr

let $5^x = u$

$3u^2 + 25u - 200 = 0$

$(u-5)(3u+40) = 0$ K1 factor

$u = 5 \;,\; u = -\dfrac{40}{3}$

$5^x = 5 \qquad 5^x = -\dfrac{40}{3}$ K1 subs both

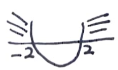
$\therefore x = 1$ J1 (no solution)

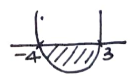
2(b) $x + 4 \leq x^2 + x < 12$

$x+4 \leq x^2 + x$ AND $x^2 + x < 12$

$x^2 - 4 \geq 0 \qquad x^2 + x - 12 < 0$ K1

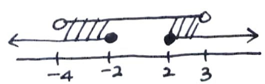
$(x+2)(x-2) \geq 0 \qquad (x+4)(x-3) < 0$





K1

$x \leq -2$ or $x \geq 2 \qquad -4 < x < 3$ J1



K1

$(-4, -2] \cup [2, 3)$ J1

③ a) $f(x) = \dfrac{1}{1 + \dfrac{1}{1 + \frac{1}{x}}}$

$= \dfrac{1}{1 + \dfrac{1}{\frac{x+1}{x}}}$

$= \dfrac{1}{1 + \frac{x}{x+1}}$ K1 simplify

$= \dfrac{x+1}{2x+1}$ J1

$f(\tfrac{1}{2}) = \dfrac{\frac{1}{2} + 1}{2(\frac{1}{2}) + 1}$ K1

$= \dfrac{3}{4}$ J1

b) $x \neq 0$ B1

$1 + \dfrac{1}{x} \neq 0 \rightarrow x \neq -1$

$1 + \dfrac{1}{1 + \frac{1}{x}} \neq 0 \rightarrow x \neq -\dfrac{1}{2}$

K1 method to find x

$\therefore$ numbers are $-\frac{1}{2}, -1, 0$

J1 2 correct

J1 all

4a) $\log_2 2x = 2\log_4 (x+4)$

$\log_2 2x = 2\,\dfrac{\log_2 (x+4)}{\log_2 4}$ (K1) change base

$= 2\,\dfrac{\log_2 (x+4)}{\log_2 2^2}$ (K1) $4 = 2^2$

$\log_2 2x = \log_2 (x+4)$ (K1) simplify (J1)

$2x = x+4$ (K1) compare

$\therefore x = 4$ (J1)

4b) $\left|\dfrac{x-3}{2x-1}\right| \geq \dfrac{1}{2}$

$\dfrac{x-3}{2x-1} \geq \dfrac{1}{2}$ OR $-\left(\dfrac{x-3}{2x-1}\right) \geq \dfrac{1}{2}$ (B1) define

$\dfrac{2x-6-2x+1}{2(2x-1)} \geq 0$ $\qquad$ $\dfrac{2x-6+2x-1}{2(2x-1)} \leq 0$ (K1) single fraction

$\dfrac{-5}{2(2x-1)} \geq 0$ $\qquad$ $\dfrac{4x-7}{2(2x-1)} \leq 0$

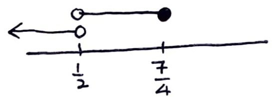
$x < \frac{1}{2}$ (J1)

T.O.S

Interv	$(-\infty, \frac{1}{2})$	$(\frac{1}{2}, \frac{7}{4}]$	$[\frac{7}{4}, \infty)$
x	-1	1	3
$4x-7$	$-$	$-$	$+$
$2x-1$	$-$	$+$	$+$
Conc	$+$	$-$	$+$

(K1)

$\frac{1}{2} < x \leq \frac{7}{4}$ (J1)



$\therefore x < \frac{1}{2}$ or $\frac{1}{2} < x \leq \frac{7}{4}$

$(-\infty, \frac{1}{2}) \cup (\frac{1}{2}, \frac{7}{4}]$ (J1)

5 $\quad f(x) = \ln(2x+1)$

a) $2x+1 > 0$

$x > -\frac{1}{2}$

$D_f : (-\frac{1}{2}, \infty)$ (B1)

$R_f : (-\infty, \infty)$ (B1)

b) $ff^{-1}(x) = x$

$\ln(2f^{-1}(x) + 1) = x$ (K1)

$2f^{-1}(x) + 1 = e^x$ (K1)

$f^{-1}(x) = \dfrac{e^x - 1}{2}$ (J1)

$D_{f^{-1}} : (-\infty, \infty)$ (B1)

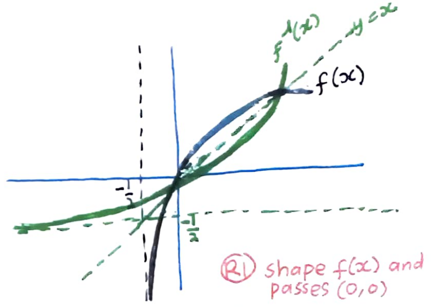
$R_{f^{-1}} : (-\frac{1}{2}, \infty)$ (B1)

$f^{-1}(x) = 0$:

$\dfrac{e^x - 1}{2} = 0$ (K1)

$e^x = 1$

$\therefore x = 0$ (J1)



(R1) shape $f(x)$ and passes $(0,0)$

(R1) shape $f^{-1}(x)$ and passes $(0,0)$

(R1) all correct (with label)

6. a) $\lim_{x \to 3^-} \frac{9-x^2}{x-3} = \lim_{x \to 3^-} \frac{(3-x)(3+x)}{x-3}$ K1

$= \lim_{x \to 3^-} -(3+x)$

K1

$= -6$ J1

$\lim_{x \to 3^+} \frac{162x-54x^2}{x^3-27} = \lim_{x \to 3^+} \frac{54x(3-x)}{(x-3)(x^2+3x+9)}$ K1 K1

$= \lim_{x \to 3^+} \frac{-54x}{x^2+3x+9}$

$= -6$ J1

Since $f(3) = \lim_{x \to 3^-} f(x) = \lim_{x \to 3^+} f(x) = -6$

$\therefore f(x)$ is continuous at $x=3$ J1

b) $\lim_{x \to \infty} \frac{5x^3+4x-9}{7-3x^3} = \lim_{x \to \infty} \frac{\frac{5x^3}{x^3} + \frac{4x}{x^3} - \frac{9}{x^3}}{\frac{7}{x^3} - \frac{3x^3}{x^3}}$ K1

$= \lim_{x \to \infty} \frac{5 + \frac{4}{x^2} - \frac{9}{x^3}}{\frac{7}{x^3} - 3}$

$= \frac{5+0-0}{0-3}$ K1

$= -\frac{5}{3}$ J1

7. a) $y = x^2 e^{-3x}$

$\frac{dy}{dx} = 2xe^{-3x} - 3x^2e^{-3x}$ K1 J1

$= e^{-3x}(2x - 3x^2)$

$\frac{d^2y}{dx^2} = -3e^{-3x}(2x-3x^2) + e^{-3x}(2-6x)$ K1 J1

$= e^{-3x}(9x^2 - 12x + 2)$ J1

b) $x^2y = \sin(2x^2+\pi)$

K1 $\left\{ 2xy + x^2\frac{dy}{dx} = 4x\cos(2x^2+\pi) \right\}$ K1 J1

K1 $\left\{ 2y + 2x\frac{dy}{dx} + 2x\frac{dy}{dx} + x^2\frac{d^2y}{dx^2} = 4\cos(2x^2+\pi) - 16x^2\sin(2x^2+\pi) \right\}$ K1

$2y + 4x\frac{dy}{dx} + x^2\frac{d^2y}{dx^2} = 4\cos(2x^2+\pi) - 16x^2\underbrace{(x^2y)}_{K1}$

$2y + 16x^4y + 4x\frac{dy}{dx} + x^2\frac{d^2y}{dx^2} = 4\cos(2x^2+\pi)$

$2y(1+8x^4) + 4x\frac{dy}{dx} + x^2\frac{d^2y}{dx^2} = 4\cos(2x^2+\pi)$ shown. J1

⑧ $y = x^2 - \frac{6}{x}$

$$\frac{dy}{dx} = 2x + \frac{6}{x^2}$$

At the stationary point,

$$\frac{dy}{dx} = 2x + \frac{6}{x^2} = 0 \quad \text{(K1)}$$

$$2x^3 = -6$$

$$\left.\begin{aligned} x &= (-3)^{1/3} = -1.442 \text{ (3dp)} \\ y &= 6.240 \end{aligned}\right\} \quad \text{(K1)}$$

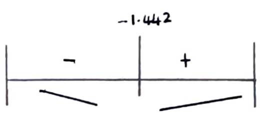
Stationary point is $(-1.442, 6.240)$ (J1)

(M1) $$\frac{d^2y}{dx^2} = 2 - 12x^{-3}$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=-1.442} = 2 - \frac{12}{(-1.442)^3} = 6.002 \ (>0) \quad \text{(K1)}$$

$\therefore (-1.442, 6.240)$ is the minimum point. (J1)

(M2)



(K1)

⑨ a) For $\lim_{x\to 2} g(x)$ to exists

$$\lim_{x\to 2^-} m^2 - 8 = \lim_{x\to 2^+} \frac{x-2}{\sqrt{2x} - 2} \quad \text{(K1)}$$

$$m^2 - 8 = \lim_{x\to 2^+} \frac{x-2}{\sqrt{2x} - 2} \cdot \frac{\sqrt{2x}+2}{\sqrt{2x}+2} \quad \text{(K1)}$$

$$= \lim_{x\to 2^+} \frac{(x-2)(\sqrt{2x}+2)}{2(x-2)} \quad \text{(K1)}$$

$$= \lim_{x\to 2^+} \frac{\sqrt{2x}+2}{2}$$

$$m^2 - 8 = 2 \quad \text{(K1)}$$

$$m^2 = 10$$

$$m = \pm\sqrt{10} \quad \text{(J1)}$$

b) For $g(x)$ is discontinuous at $x = 8$

$$\lim_{x\to 8^-} \frac{x-2}{\sqrt{2x} - 2} \neq \lim_{x\to 8^+} \frac{|8-x|}{x-8} + k \quad \text{(K1)}$$

$$\frac{6}{2} \neq \lim_{x\to 8^+} \frac{x-8}{x-8} + k \quad \text{(K1)}$$

$$3 \neq 1 + k$$

$$k \neq 2 \quad \text{(J1)}$$

(10) a). Volume of the cylinder,

$$V = \pi r^2 h$$

$$= \pi \left(\sqrt{a^2-h^2}\right)^2 (2h) \quad \text{K1}$$

$$= 2\pi (a^2-h^2) h \quad \text{J1}$$

Hence,

$$V = 2\pi a^2 h - 2\pi h^3$$

$$\frac{dV}{dh} = 2\pi a^2 - 6\pi h^2 \quad \text{K1} \quad \text{J1}$$

$$\frac{d^2V}{dx^2} = -12\pi h \quad \text{J1}$$

b) $\frac{dV}{dh} = 0$

$$2\pi a^2 - 6\pi h^2 = 0 \quad \text{K1}$$

$$\therefore h = \frac{a}{\sqrt{3}} \quad \text{K1} \quad \text{J1}$$

$$\left.\frac{d^2V}{dh^2}\right|_{h=\frac{a}{\sqrt{3}}} = -12\pi\left(\frac{a}{\sqrt{3}}\right) < 0 \text{ because } a > 0$$

K1 J1

$$V_{max} = 2\pi\left[a^2 - \left(\frac{a}{\sqrt{3}}\right)^2\right]\left[\frac{a}{\sqrt{3}}\right] \quad \text{K1}$$

$$= 2\pi\left(\frac{2a^2}{3}\right)\left(\frac{a}{\sqrt{3}}\right)$$

$$= \frac{4\pi a^3}{3\sqrt{3}}$$

$$= \frac{4\sqrt{3}\pi a^3}{9} \quad \text{J1}$$

when $a = 3$

$$V = \frac{4\sqrt{3}\pi (3)^3}{9} = 12\sqrt{3}\pi$$

K1 J1

c) Volume of sphere $= \frac{4}{3}\pi a^3$

Volume of sphere : Max. volume of cylinder

$$\frac{4}{3}\pi a^3 : \frac{4\sqrt{3}\pi a^3}{9} \quad \text{K1}$$

$$\frac{\frac{4}{3}\pi a^3}{\frac{4\sqrt{3}\pi a^3}{9}} : 1 \quad \text{K1}$$

$$\sqrt{3} : 1 \quad \text{J1}$$